

Unit 5

1. The Montreal Protocol

The Montreal Protocol on Substances that Deplete the Ozone Layer is the landmark multilateral environmental agreement that regulates the production and consumption of nearly 100 man-made chemicals referred to as ozone depleting substances (ODS). When released to the atmosphere, those chemicals damage the stratospheric ozone layer, Earth's protective shield that protects humans and the environment from harmful levels of ultraviolet radiation from the sun. Adopted on 16 September 1987, the Protocol is to date one of the rare treaties to achieve universal ratification.

The Montreal Protocol phases down the consumption and production of the different ODS in a step-wise manner, with different timetables for developed and developing countries (referred to as "Article 5 countries"). Under this treaty, all parties have specific responsibilities related to the phase out of the different groups of ODS, control of ODS trade, annual reporting of data, national licensing systems to control ODS imports and exports, and other matters. Developing and developed countries have equal but differentiated responsibilities, but most importantly, both groups of countries have binding, time-targeted and measurable commitments.

The Protocol includes provisions related to Control Measures (Article 2), Calculation of control levels (Article 3), Control of trade with non-Parties (Article 4), Special situation of developing countries (Article 5), Reporting of data (Article 7), Non-compliance (Article 8), Technical assistance (Article 10), as well as other topics. The substances controlled by the treaty are listed in Annexes A (CFCs, halons), B (other fully halogenated CFCs, carbon tetrachloride, methyl chloroform), C (HCFCs), E (methyl bromide) and F (HFCs).

The treaty evolves over time in light of new scientific, technical and economic developments, and it continues to be amended and adjusted. The Meeting of the Parties is the governance body for the treaty, with technical support provided by an Open-ended Working Group, both of which meet on an annual basis. The Parties are assisted by the Ozone Secretariat, which is based at UN Environment Programme headquarters in Nairobi, Kenya.

1.1. The Multilateral Fund

The Multilateral Fund for the Implementation of the Montreal Protocol was established in 1991 under Article 10 of the treaty. The Fund's objective is to provide financial and technical assistance to developing country parties to the Montreal Protocol whose annual per capita consumption and production of ODS is less than 0.3 kg to comply with the control measures of the Protocol.

The Multilateral Fund's activities are implemented by four international agencies - UN Environment Programme (UNEP), UN Development Programme (UNDP), UN Industrial Development Organisation (UNIDO) and the World Bank - as well as bilateral agencies of non-Article 5 countries.

Responsibility for overseeing the operation of the Fund rests with the Executive Committee, which comprises seven members each from Article 5 countries and non-Article 5 countries. The Committee is assisted by the Multilateral Fund Secretariat, which is based in Montreal. Since its inception, the Multilateral Fund has supported over 8,600 projects including industrial conversion, technical assistance, training and capacity building worth over US\$3.9 billion.

Throughout the implementation of the Montreal Protocol, developing countries have demonstrated that, with the right kind of assistance, they are willing, ready and able to be full partners in global

efforts to protect the environment. In fact, many developing countries have exceeded the reduction targets for phasing out ODS, with the support of the Multilateral Fund.

Phase out of HCFCs – the Montreal Amendment

Hydrochlorofluorocarbons (HCFCs) are gases used worldwide in refrigeration, air-conditioning and foam applications, but they are being phased out under the Montreal Protocol since deplete the ozone layer. HCFCs are both ODS and powerful greenhouse gases: the most commonly used HCFC is nearly 2,000 times more potent than carbon dioxide in terms of its global warming potential (GWP). Recognizing the potential benefits to the Earth's climate, in September 2007 the Parties decided to accelerate their schedule to phase out HCFCs. Developed countries have been reducing their consumption of HCFCs and will completely phase them out by 2020. Developing countries agreed to start their phase-out process in 2013 and are now following a stepwise reduction until the complete phase-out of HCFCs by 2030.

In Article 5 countries, this HCFC phase-out is in full swing, with support from the Multilateral Fund for the implementation of multi-stage HCFC Phase out Management Plans (HPMPs), investment projects and capacity building activities. Throughout this process, the Parties are encouraging all countries to promote the selection of alternatives to HCFCs that minimize environmental impacts, in particular impacts on climate, as well as meeting other health, safety and economic considerations. For the climate consideration, this means taking global-warming potential, energy use and other relevant factors into account. For refrigeration and air conditioning, this means optimizing refrigerants, equipment, servicing practices, recovery, recycling and disposal at end of life.

Phase down of HFCs – the Kigali Amendment

Another group of substances, hydrofluorocarbons (HFCs), were introduced as non-ozone depleting alternatives to support the timely phase-out of CFCs and HCFCs. HFCs are now widespread in air conditioners, refrigerators, aerosols, foams and other products. While these chemicals do not deplete the stratospheric ozone layer, some of them have high GWPs ranging from 12 to 14,000. Overall HFC emissions are growing at a rate of 8% per year and annual emissions are projected to rise to 7-19% of global CO₂ emissions by 2050. Uncontrolled growth in HFC emissions, therefore, challenges efforts to keep global temperature rise at or below 2°C this century. Urgent action on HFCs is needed to protect the climate system.

The Parties to the Montreal Protocol reached an agreement at their 28th Meeting of the Parties on 15 October 2016 in Kigali, Rwanda to phase down HFCs. Countries agreed to add HFCs to the list of controlled substances and approved a timeline for their gradual reduction by 80-85 per cent by the late 2040s. The first reductions by developed countries are expected in 2019. Developing countries will follow with a freeze of HFC consumption levels in 2024 and in 2028 for some nations.

The issue has been under negotiation by the Parties since 2009 and the successful agreement on the Kigali Amendment (Decision XXVIII/1 and accompanying Decision XXVIII/2) continues the historic legacy of the Montreal Protocol. The Kigali Amendment will enter into force on 1 January 2019 for those countries that have ratified the amendment.

The pathway to implement the HFC phase down is to reduce dependency on high-GWP alternatives and increase the adoption of low-GWP, energy-efficient technologies as part of the HCFC phase-out process under the Montreal Protocol. Such a “smart approach” can achieve the Montreal Protocol's objective of eliminating HCFCs while at the same time achieving energy efficiency gains and CO₂ emissions reduction — a “climate co-benefit.”

2. Kyoto Protocol

The Kyoto Protocol was adopted on 11 December 1997. Owing to a complex ratification process, it entered into force on 16 February 2005. Currently, there are 192 Parties to the Kyoto Protocol. The Kyoto Protocol recognized that developed countries are principally responsible for the current high levels of GHG emissions in the atmosphere as a result of more than 150 years of industrial activity. As such, the protocol placed a heavier burden on developed nations than less-developed nations. The Kyoto Protocol mandated that 37 industrialized nations plus the EU cut their GHG emissions.

In short, the Kyoto Protocol operationalizes the United Nations Framework Convention on Climate Change by committing industrialized countries and economies in transition to limit and reduce greenhouse gases (GHG) emissions in accordance with agreed individual targets. The Convention itself only asks those countries to adopt policies and measures on mitigation and to report periodically.

The Kyoto Protocol is based on the principles and provisions of the Convention and follows its annex-based structure. It only binds developed countries, and places a heavier burden on them under the principle of “common but differentiated responsibility and respective capabilities”, because it recognizes that they are largely responsible for the current high levels of GHG emissions in the atmosphere.

In its Annex B, the Kyoto Protocol sets binding emission reduction targets for 37 industrialized countries and economies in transition and the European Union. Overall, these targets add up to an average 5 per cent emission reduction compared to 1990 levels over the five year period 2008–2012 (the first commitment period).

2.1. Kyoto Mechanisms

These mechanisms ideally encourage GHG abatement to start where it is most cost-effective, for example, in the developing world. It does not matter where emissions are reduced, as long as they are removed from the atmosphere. This has the parallel benefits of stimulating green investment in developing countries and including the private sector in this endeavour to cut and hold steady GHG emissions at a safe level. It also makes leap-frogging—that is, the possibility of skipping the use of older, dirtier technology for newer, cleaner infrastructure and systems, with obvious longer-term benefits—more economical. The Kyoto Protocol established three different mechanisms to enable countries additional ways to meet their emission-limitation target. The three mechanisms are:

The International Emissions Trading mechanism: Countries that have excess emission units permitted to them but not used can engage in carbon trading and sell these units to countries over their target.

The Clean Development mechanism: Countries with emission reducing or limiting commitments may implement emission-reducing projects in developing countries to earn certified emission reduction credits.

The Joint Implementation mechanism: Countries with emission reducing or limiting commitments to earn emission reducing units from a project in another party.

Some of the principal concepts of the Kyoto Protocol are:

- Binding commitments for the Annex I Parties. The main feature of the Protocol is that it established legally binding commitments to reduce emissions of greenhouse gases for Annex I Parties. The commitments were based on the Berlin Mandate, which was a part of UNFCCC negotiations leading up to the Protocol.

- Implementation. In order to meet the objectives of the Protocol, Annex I Parties are required to prepare policies and measures for the reduction of greenhouse gases in their respective countries. In addition, they are required to increase the absorption of these gases and utilize all mechanisms available, such as joint implementation, the clean development mechanism and emissions trading, in order to be rewarded with credits that would allow more greenhouse gas emissions at home.
- Minimizing Impacts on Developing Countries by establishing an adaptation fund for climate change.
- Accounting, Reporting and Review in order to ensure the integrity of the Protocol.
- Compliance. Establishing a Compliance Committee to enforce compliance with the commitments under the Protocol.

2.2. Chronology

1992 – The UN Conference on the Environment and Development is held in Rio de Janeiro. It results in the Framework Convention on Climate Change ("FCCC" or "UNFCCC") among other agreements.

1995 – Parties to the UNFCCC meet in Berlin (the 1st Conference of Parties (COP) to the UNFCCC) to outline specific targets on emissions.

1997 – In December the parties conclude the Kyoto Protocol in Kyoto, Japan, in which they agree to the broad outlines of emissions targets.

2004 – Russia and Canada ratify the Kyoto Protocol to the UNFCCC bringing the treaty into effect on 16 February 2005.

2011 – Canada became the first signatory to announce its withdrawal from the Kyoto Protocol.[22]

2012 – On 31 December 2012, the first commitment period under the Protocol expired.

2.3. Kyoto Protocol - Targets for the first commitment period

The targets for the first commitment period of the Kyoto Protocol cover emissions of the six main greenhouse gases, namely:

- Carbon dioxide (CO₂);
- Methane (CH₄);
- Nitrous oxide (N₂O);
- Hydrofluorocarbons (HFCs);
- Perfluorocarbons (PFCs); and
- Sulphur hexafluoride (SF₆)

The maximum amount of emissions (measured as the equivalent in carbon dioxide) that a Party may emit over a commitment period in order to comply with its emissions target is known as a Party's assigned amount. The individual targets for Annex I Parties are listed in the Kyoto Protocol's Annex B.

In Doha, Qatar, on 8 December 2012, **the Doha Amendment** to the Kyoto Protocol was adopted for a second commitment period, starting in 2013 and lasting until 2020. As of 28 October 2020, 147 Parties deposited their instrument of acceptance, therefore the threshold of 144 instruments of acceptance for entry into force of the Doha Amendment was achieved. The amendment entered into force on 31 December 2020. The amendment includes:

- New commitments for Annex I Parties to the Kyoto Protocol who agreed to take on commitments in a second commitment period from 1 January 2013 to 31 December 2020;

- A revised list of GHG to be reported on by Parties in the second commitment period; and
- Amendments to several articles of the Kyoto Protocol which specifically referenced issues pertaining to the first commitment period and which needed to be updated for the second commitment period.

2.4. Monitoring

The Kyoto Protocol also established a rigorous monitoring, review and verification system, as well as a compliance system to ensure transparency and hold Parties to account. Under the Protocol, countries' actual emissions have to be monitored and precise records have to be kept of the trades carried out.

- **Registry systems** track and record transactions by Parties under the mechanisms. The UN Climate Change Secretariat, based in Bonn, Germany, keeps an international transaction log to verify that transactions are consistent with the rules of the Protocol.
- **Reporting** is done by Parties by submitting annual emission inventories and national reports under the Protocol at regular intervals.
- A **compliance system** ensures that Parties are meeting their commitments and helps them to meet their commitments if they have problems doing so.
- The Kyoto Protocol, like the Convention, is also designed to assist countries in **adapting to the adverse effects of climate change**. It facilitates the development and deployment of technologies that can help increase resilience to the impacts of climate change.
- The **Adaptation Fund** was established to finance adaptation projects and programmes in developing countries that are Parties to the Kyoto Protocol. In the first commitment period, the Fund was financed mainly with a share of proceeds from CDM project activities. In Doha, in 2012, it was decided that for the second commitment period, international emissions trading and joint implementation would also provide the Adaptation Fund with a 2 percent share of proceeds.

3. Clean Development Mechanism

The Clean Development Mechanism (CDM), defined in Article 12 of the Protocol, allows a country with an emission-reduction or emission-limitation commitment under the Kyoto Protocol (Annex B Party) to implement an emission-reduction project in developing countries. Such projects can earn saleable certified emission reduction (CER) credits, each equivalent to one tonne of CO₂, which can be counted towards meeting Kyoto targets.

The mechanism is seen by many as a trailblazer. It is the first global, environmental investment and credit scheme of its kind, providing a standardized emissions offset instrument, CERs. A CDM project activity might involve, for example, a rural electrification project using solar panels or the installation of more energy-efficient boilers. The mechanism stimulates sustainable development and emission reductions, while giving industrialized countries some flexibility in how they meet their emission reduction or limitation targets.

A CDM project must provide emission reductions that are additional to what would otherwise have occurred. The projects must qualify through a rigorous and public registration and issuance process. Approval is given by the Designated National Authorities. Public funding for CDM project activities must not result in the diversion of official development assistance.

The mechanism is overseen by the CDM Executive Board, answerable ultimately to the countries that have ratified the Kyoto Protocol. Operational since the beginning of 2006, the mechanism has already registered more than 1,650 projects and is anticipated to produce CERs amounting to more than 2.9 billion tonnes of CO₂ equivalent in the first commitment period of the Kyoto Protocol, 2008–2012.

4. Paris Agreement

The Paris Agreement is a legally binding international treaty on climate change. It was adopted by 196 Parties at the UN Climate Change Conference (COP21) in Paris, France, on 12 December 2015. It entered into force on 4 November 2016.

Its overarching goal is to hold “the increase in the global average temperature to well below 2°C above pre-industrial levels” and pursue efforts “to limit the temperature increase to 1.5°C above pre-industrial levels.”

However, in recent years, world leaders have stressed the need to limit global warming to 1.5°C by the end of this century. That’s because the UN’s Intergovernmental Panel on Climate Change indicates that crossing the 1.5°C threshold risks unleashing far more severe climate change impacts, including more frequent and severe droughts, heatwaves and rainfall. To limit global warming to 1.5°C, greenhouse gas emissions must peak before 2025 at the latest and decline 43% by 2030.

The Paris Agreement is a landmark in the multilateral climate change process because, for the first time, a binding agreement brings all nations together to combat climate change and adapt to its effects.

4.1. Implementation

Implementation of the Paris Agreement requires economic and social transformation, based on the best available science. The Paris Agreement works on a five-year cycle of increasingly ambitious climate action -- or, ratcheting up -- carried out by countries. Since 2020, countries have been submitting their national climate action plans, known as nationally determined contributions (NDCs). Each successive NDC is meant to reflect an increasingly higher degree of ambition compared to the previous version.

Recognizing that accelerated action is required to limit global warming to 1.5°C, the COP27 cover decision requests Parties to revisit and strengthen the 2030 targets in their NDCs to align with the Paris Agreement temperature goal by the end of 2023, taking into account different national circumstances.

- **NDCs:** countries communicate actions they will take to reduce their greenhouse gas emissions in order to reach the goals of the Paris Agreement. Countries also communicate in their NDCs actions they will take to build resilience to adapt to the impacts of climate change.
- **Long Term Strategies:** To better frame the efforts towards the long-term goal, the Paris Agreement invites countries to formulate and submit long-term low greenhouse gas emission development strategies (LT-LEDS). LT-LEDS provide the long-term horizon to the NDCs. Unlike NDCs, they are not mandatory. Nevertheless, they place the NDCs into the context of countries’ long-term planning and development priorities, providing a vision and direction for future development.

5. Nationally Determined Contributions

Nationally determined contributions (NDCs) are at the heart of the Paris Agreement and the achievement of its long-term goals. NDCs embody efforts by each country to reduce national emissions and adapt to the impacts of climate change. The Paris Agreement requires each Party to prepare, communicate and maintain successive nationally determined contributions (NDCs) that it intends to achieve. Parties shall pursue domestic mitigation measures, with the aim of achieving the objectives

of such contributions. The Paris Agreement requests each country to outline and communicate their post-2020 climate actions, known as their NDCs.

Together, these climate actions determine whether the world achieves the long-term goals of the Paris Agreement and to reach global peaking of greenhouse gas (GHG) emissions as soon as possible and to undertake rapid reductions thereafter in accordance with best available science, so as to achieve a balance between anthropogenic emissions by sources and removals by sinks of GHGs in the second half of this century. It is understood that the peaking of emissions will take longer for developing country Parties, and that emission reductions are undertaken on the basis of equity, and in the context of sustainable development and efforts to eradicate poverty, which are critical development priorities for many developing countries.

An NDC is a government obligation under the Paris Agreement, one or more national ministries will generally lead its development. But for NDCs to work, they need to be widely understood and used by businesses, civil society, academia and ordinary citizens. Each has roles to play, which is why many governments invite different constituencies to take part in defining NDC priorities.

5.1. NDCs of India

- I. To put forward and further propagate a healthy and sustainable way of living based on traditions and values of conservation and moderation, including through a mass movement for 'LIFE' – 'Lifestyle for Environment' as a key to combating climate change.
- II. To adopt a climate friendly and a cleaner path than the one followed hitherto by others at corresponding level of economic development.
- III. To reduce Emissions Intensity of its GDP by 45 percent by 2030, from 2005 level.
- IV. To achieve about 50 percent cumulative electric power installed capacity from non-fossil fuel-based energy resources by 2030, with the help of transfer of technology and low-cost international finance including from Green Climate Fund (GCF).
- V. To create an additional carbon sink of 2.5 to 3 billion tonnes of CO₂ equivalent through additional forest and tree cover by 2030.
- VI. To better adapt to climate change by enhancing investments in development programmes in sectors vulnerable to climate change, particularly agriculture, water resources, Himalayan region, coastal regions, health and disaster management.
- VII. To mobilize domestic and new & additional funds from developed countries to implement the above mitigation and adaptation actions in view of the resource required and the resource gap.
- VIII. To build capacities, create domestic framework and international architecture for quick diffusion of cutting-edge climate technology in India and for joint collaborative R&D for such future technologies.

6. Carbon Credits/Carbon Offsets

A carbon credit or offset credit is a transferrable instrument certified by governments or independent certification bodies to represent an emission reduction that can then be bought or sold. Both offsets and credits are measured in tonnes of carbon dioxide-equivalent (CO₂e). One ton of carbon offset or credit represents the reduction or removal of one ton of carbon dioxide or its equivalent in other greenhouse gases. Carbon credits, also known as carbon offsets, are permits that allow the owner to emit a certain amount of carbon dioxide or other greenhouse gases. One credit permits the emission of one ton of carbon dioxide or the equivalent in other greenhouse gases.

A carbon credit or offset credit is a transferrable instrument certified by governments or independent certification bodies to represent an emission reduction of one metric ton of CO₂, or an equivalent

amount of other greenhouse gases (GHGs). Carbon offsets and credits, along with carbon taxes and subsidies, are all forms of carbon pricing. Historically, the concepts of offsets and credits have been intertwined. Both offsets and credits can move amongst the various markets they are traded in. Some programs use other terms for these concepts. Voluntary emissions reduction (VER) refers to a carbon offset that is exchanged in the over-the-counter or voluntary market for credits. Certified emissions reduction (CER) credits are created through a regulatory framework such as the Clean Development Mechanism.

Carbon offset and credit terminology is continuing to evolve. At COP27 negotiators agreed to define offsets and credits issued under Article 6 of the Paris Agreement "mitigation contributions", as a means of discouraging carbon neutrality claims by buyers. Certification organizations such as the Gold Standard even have detailed guidance on how emission reduction projects should be described.

The carbon credit is half of a so-called cap-and-trade program. Companies that pollute are awarded credits that allow them to continue to pollute up to a certain limit, which is reduced periodically. Meanwhile, the company may sell any unneeded credits to another company that needs them. Private companies are thus doubly incentivized to reduce greenhouse emissions. First, they must spend money on extra credits if their emissions exceed the cap. Second, they can make money by reducing their emissions and selling their excess allowances.

Proponents of the carbon credit system say that it leads to measurable, verifiable emission reductions from certified climate action projects, and that these projects reduce, remove, or avoid greenhouse gas (GHG) emissions.

The ultimate goal of carbon credits is to reduce the emission of greenhouse gases into the atmosphere. As noted, a carbon credit represents the right to emit greenhouse gases equivalent to one ton of carbon dioxide. According to the Environmental Defense Fund, that is the equivalent of a 2,400-mile drive in terms of carbon dioxide emissions.

Companies or nations are allotted a certain number of credits and may trade them to help balance total worldwide emissions. "Since carbon dioxide is the principal greenhouse gas," the United Nations notes, "people speak simply of trading in carbon."

6.1. Carbon Credit important features

- **Vintage:** The vintage is the year in which the carbon emissions reduction project generates the carbon offset credit. This is usually done once a third party verifies the project. This can be done by a validation-verification body, a designated operational entity, or other accredited third party reviewers. However, there is a practice called "Forward Crediting" employed by a limited number of programs, whereby credits may be issued for projected emission reductions that the project developer anticipates. This practice risks over-issuing credits if the project does not realize its estimated impact, and allows credit buyers to claim emission reductions in the present for activities that have not yet occurred.
- **Project type:** A variety of projects can be used to reduce GHG emissions. These can include land-use (e.g. improved forestry management), methane capture, biomass sequestration, renewable energy, industrial energy efficiency, and more.
- **Co-benefits:** Beyond reducing greenhouse gas emissions, projects may provide benefits such as ecosystem services or economic opportunities for communities near the project site. These project benefits are termed "co-benefits". For example, projects that reduce agricultural greenhouse gas emissions may improve water quality by reducing fertilizer usage that results in run-off and may contaminate water.

- **Certification regime:** The certification regime describes the systems and procedures that are used to certify and register carbon offsets and credits. These vary in terms of governance and accounting practices, project eligibility, environmental integrity and sustainable development requirements, and Monitoring, Reporting and Verification (MRV) procedures.
- **Carbon retirement:** Offset credit holders must “retire” carbon offset credits in order to claim their associated GHG reductions towards a specific GHG reduction goal. In the voluntary market, carbon offset registries define the manner in which retirement happens. Once an offset credit is retired, it cannot be transferred or used (meaning it is effectively taken out of circulation).[28] Voluntary purchasers can also offset their carbon emissions by purchasing carbon allowances from legally mandated cap-and-trade programs such as the Regional Greenhouse Gas Initiative or the European Emissions Trading Scheme.

7. Prototype Carbon Funds

Recognizing that global warming will have the most impact on its borrowing client countries, the World Bank approved the establishment of the Prototype Carbon Fund (PCF). The PCF is intended to invest in projects that will produce high quality greenhouse gas emission reductions that could be registered with the United Nations Framework Convention on Climate Change (UNFCCC) for the purposes of the Kyoto Protocol. To increase the likelihood that the reductions will be recognized by the Parties to the UNFCCC, independent experts will follow validation, verification and certification procedures that respond to UNFCCC rules as they develop.

The PCF will pilot production of emission reductions within the framework of Joint Implementation (JI) and the Clean Development Mechanism (CDM). The PCF will invest contributions made by companies and governments in projects designed to produce emission reductions fully consistent with the Kyoto Protocol and the emerging framework for JI and the CDM. Contributors, or "Participants" in the PCF, will receive a pro rata share of the emission reductions, verified and certified in accordance with agreements reached with the respective countries "hosting" the projects.

The prototype carbon fund (PCF) is a public-private partnership whose mission is to pioneer a market for project-based greenhouse gas emission reductions within the framework of the Kyoto protocol to the United Nations Framework Convention on Climate Change (UNFCCC). PCF seeks to show how project-based greenhouse gas emission reduction transactions can lower the cost of compliance with Kyoto, promote sustainable development, and mobilize new resources for Bank clients. Recognizing the global environmental benefits of emissions reductions regardless of location, Kyoto allows industrialized countries and firms to offset certain obligations through the purchase of lower-cost emission reductions (ERs) in developing and in-transition countries. This review identifies Bank comparative advantage-related issues, as well as the issues raised by competition between the Bank and International Finance Corporation (IFC) in carbon finance.